

Graphs of trigonometric functions

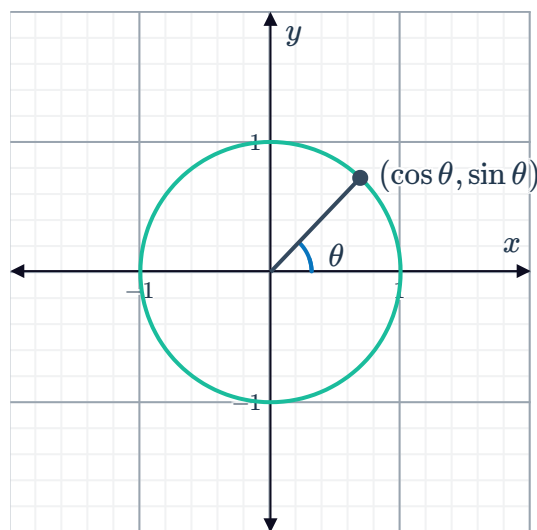


1 The value of $\sin \theta$ can be represented on the given xy -plane.

a Consider the curve $y = \sin \theta$:

Complete the following:

- i A rotation of 380° has the same rotation as the acute angle with measure $\square$. So $\sin 380^\circ = \sin \square$.
- ii A rotation of 480° has the same rotation as the obtuse angle with measure $\square$. So $\sin 480^\circ = \sin \square$.
- iii A rotation of 580° has the same rotation as the reflex angle with measure $\square$. So $\sin 580^\circ = \sin \square^\circ$.

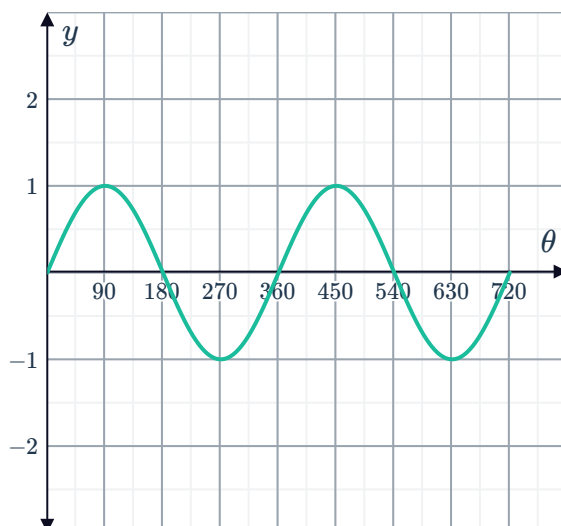


b Consider the graph of $y = \sin \theta$ for $0 \leq \theta \leq 720^\circ$:

Describe the nature of $y = \sin \theta$.

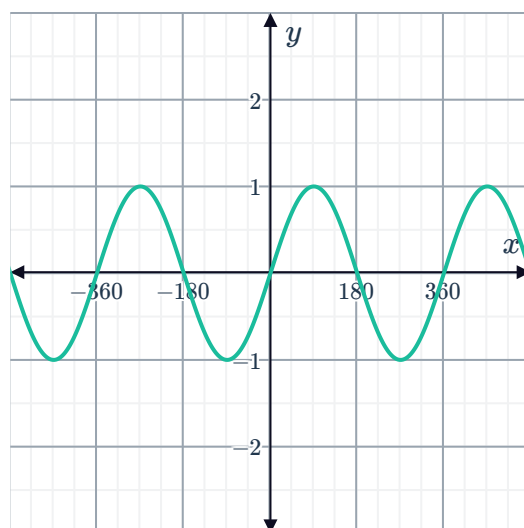
c The number of degrees it takes for the curve to complete a full cycle is called the period of the function.

Determine the period of $y = \sin \theta$.



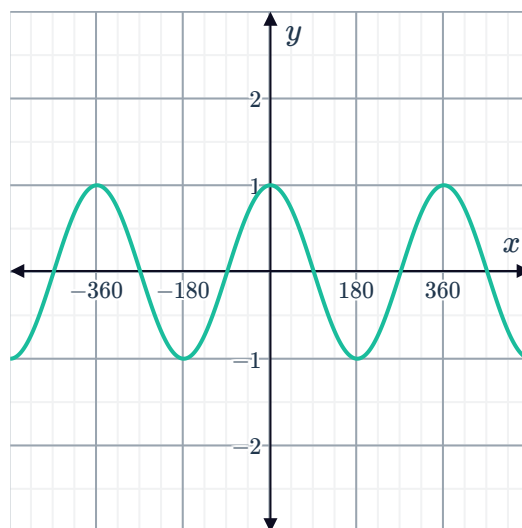
2 Consider the curve $y = \sin x$ and state whether the following statements are true or false:

- a The graph of $y = \sin x$ is periodic.
- b As x approaches infinity, the height of the graph approaches infinity.
- c The graph of $y = \sin x$ is increasing between $x = (-90)^\circ$ and $x = 0^\circ$.
- d The graph of $y = \sin x$ is symmetric about the line $x = 0$.
- e The graph of $y = \sin x$ is symmetric with respect to the origin.
- f The y -values of the graph repeat after a period of 360° .



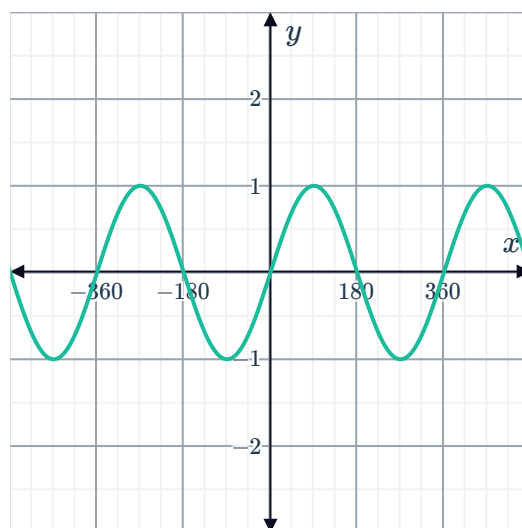
3 Consider the curve $y = \cos x$ and state whether the following statements are true or false:

- a The graph of $y = \cos x$ is cyclic.
- b As x approaches infinity, the height of the graph approaches infinity.
- c The graph of $y = \cos x$ is increasing between $x = (-180)^\circ$ and $x = (-90)^\circ$.
- d The graph of $y = \cos x$ has reflective symmetry across the line $x = 0$.
- e The graph of $y = \cos x$ has rotational symmetry with respect to the origin.
- f The y -values of the graph repeat after a period of 180° .



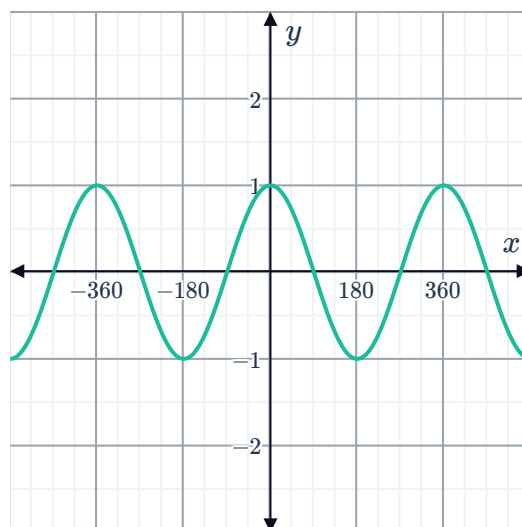
4 Consider the curve $y = \sin x$:

- a State the coordinates of the y -intercept.
- b Find the maximum y -value.
- c Find the minimum y -value.



5 Consider the curve $y = \cos x$:

- a State the coordinates of the y -intercept.
- b Find the maximum y -value.
- c Find the minimum y -value.



a Complete the table of values:



b Sketch the graph of $y = \cos x$.

x	0°	90°	180°	270°	360°
$\cos x$					

7 Consider the equation $y = \sin x$.

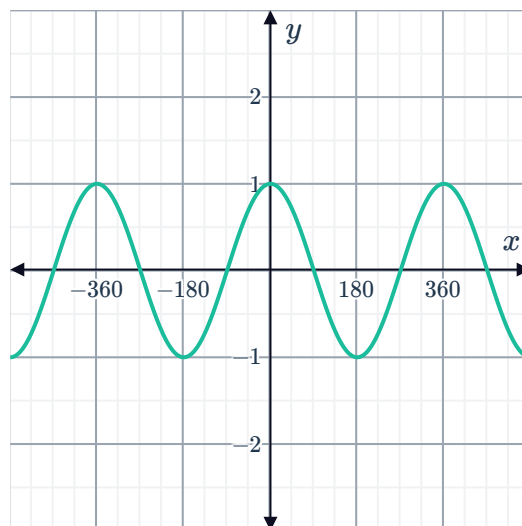
a Complete the table of values:

x	0°	90°	180°	270°	360°
$\sin x$					

b Sketch the graph of $y = \sin x$.

8 Consider the graph of $y = \cos x$:

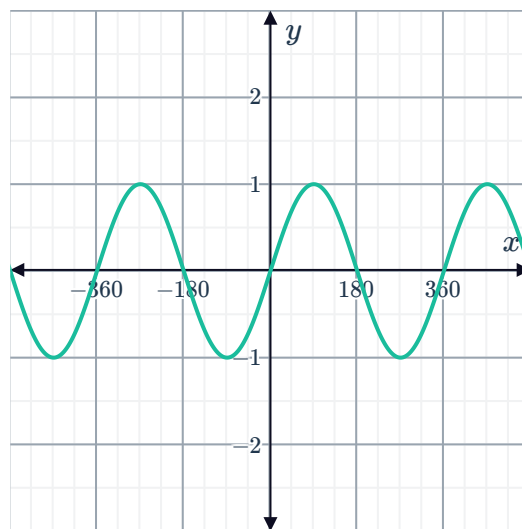
- a** Is $\cos 162^\circ$ positive or negative?
- b** In which quadrant does an angle with measure 162° lie in?



9 Consider the graph of $y = \sin x$:

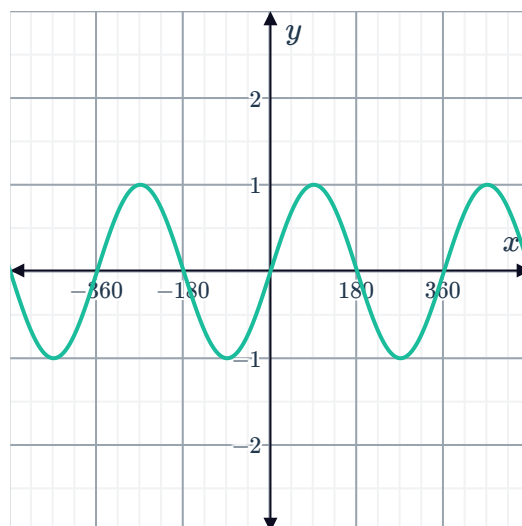


- a Is $\sin 200^\circ$ positive or negative?
- b In which quadrant does an angle with measure 200° lie in?



10 Consider the curve $y = \sin x$:

- a If one cycle of the graph of $y = \sin x$ starts at $x = 0^\circ$, when does the next cycle start?
- b Find the x -value of the x -intercept in the following regions:
 - i $0^\circ < x < 360^\circ$
 - ii $(-360)^\circ < x < 0^\circ$
- c True or false: As x approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.
- d State whether the graph of $y = \sin x$ increases or decreases in the following regions:
 - i $90^\circ < x < 270^\circ$
 - ii $(-270)^\circ < x < (-90)^\circ$
 - iii $(-450)^\circ < x < (-270)^\circ$
 - iv $(-90)^\circ < x < 90^\circ$
 - v $270^\circ < x < 450^\circ$



11 Consider the curve $y = \cos x$:



- a If one cycle of the graph of $y = \cos x$ starts at $x = (-90)^\circ$, when does the next cycle start?
- b Find the x -values of the x -intercepts in the following regions:
 - i $0^\circ < x < 360^\circ$
 - ii $(-360)^\circ < x < 0^\circ$
- c True or false: As x approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.
- d State whether the graph of $y = \cos x$ increases or decreases in the following regions:
 - i $0^\circ < x < 180^\circ$
 - ii $(-180)^\circ < x < 0^\circ$
 - iii $180^\circ < x < 360^\circ$
 - iv $(-360)^\circ < x < (-180)^\circ$

